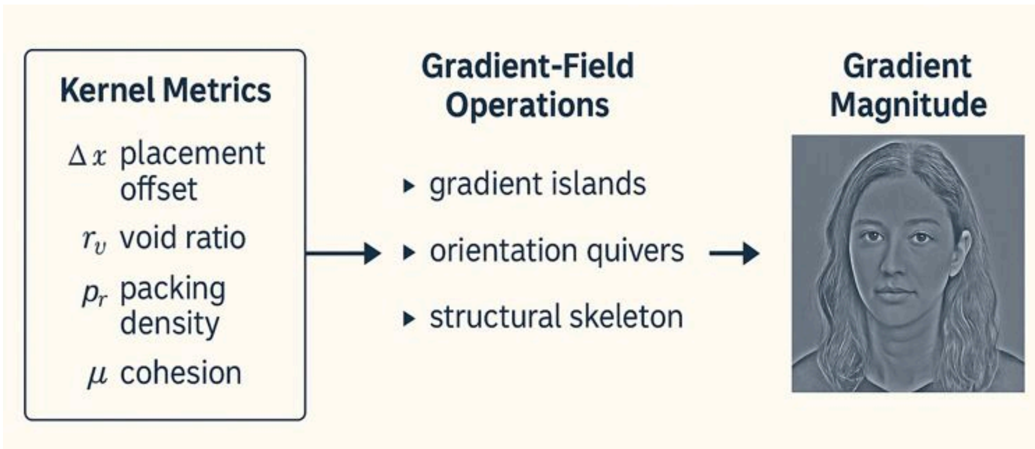
Precise Mapping: Kernel Metrics → Gradient-Field Operations

A Deterministic Framework for Measuring Spatial Priors in Images



Executive Summary

This framework defines a deterministic, reproducible method for measuring **spatial priors** in images using only **gradient-field operations**. Instead of interpreting scenes semantically, the system treats every image as a **force field**: a distribution of pressure, void, flow, and structural mass. These patterns reveal where generative models' inductive biases live, not in meaning, but in **geometry**

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Seven kernel metrics: Δx (placement offset), r_v (void ratio), ρ_r (packing density), μ (cohesion), x (peripheral pull), θ (orientation stability), and ds (structural thickness) are each defined in two ways:

1. **Compositional intuition** (the spatial phenomenon being measured)
2. **Gradient-field instantiation** (a precise, contrast-invariant operation on $|\nabla I|$)

This dual structure keeps the system interpretable while remaining fully model-agnostic. No training data, saliency methods, or internal embeddings are required. Every metric arises from first-order structure in the image itself.

By conducting all analysis in gradient space, the method becomes invariant to color, exposure, and style while remaining sensitive to **the geometric decisions that define composition**. This enables consistent measurement of model behavior, identification of persistent priors (such as center bias and void compression), and detection of **forbidden compositional zones** that models rarely reach, even under explicit prompting.

In short: **These metrics convert raw pixels into a stable map of spatial intention.** The result is a unified, interpretable substrate for evaluating, steering, and comparing generative systems.

How to Read This Document:
Each kernel metric in the Visual Thinking Lens is a two-part object:

1. A geometric intuition

- what the metric describes in compositional terms
- why it matters for spatial priors

2. A gradient-field instantiation

- how the metric is computed directly from $|\nabla I|$
- why this operation is contrast-invariant and model-agnostic

This document proceeds metric-by-metric, pairing intuition with explicit computation.

All definitions follow the same structure: **Field** → **Mask** → **Statistic** → **Interpretation**
- **Field:** Sobel-derived gradient magnitude and orientation
- Mask: adaptive percentile thresholds (low for voids, high for structure)
- Statistic: centroid, entropy, density, histogram, skeleton ratio
- Interpretation: compositional priors (void width, tension, off-center pull, fragmentation)

This structure makes the system reproducible without hidden heuristics. Every abstraction has a measurable grounding, and every gradient operation maps back to a compositional behavior.

Readers can skim:

- **Intuition:** if they want conceptual grounding
- **Mathematical Appendix:** if they want formal precision
- **Reviewer FAQ:** if they want methodological justification

→ See companion Notebook for Gradient-Based Compositional Masking for Kernel Metrics

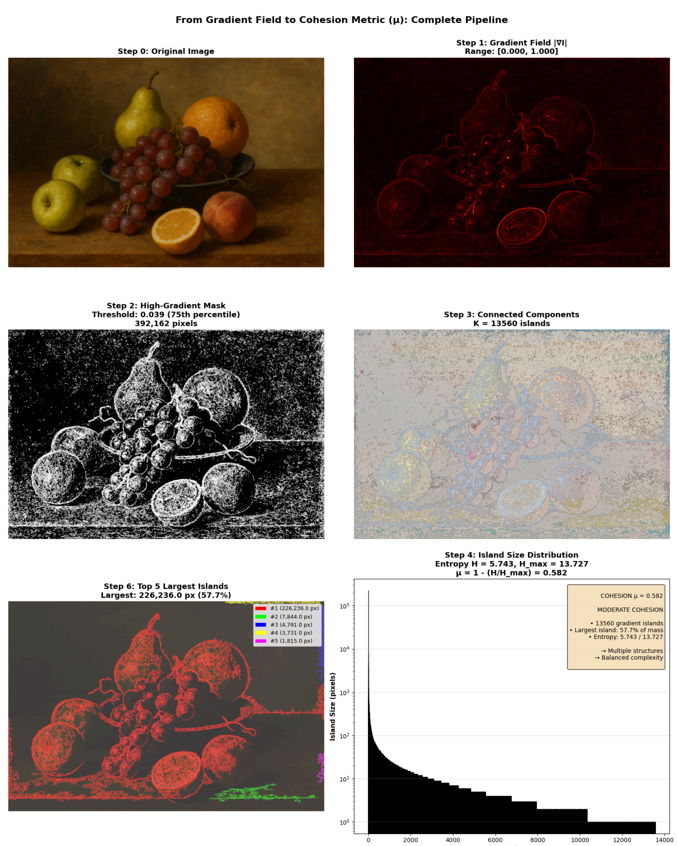

Fig 1 example:
Step 0: Original
Step 1: Gradient field $|\nabla I|$ (the foundation)
Step 2: Thresholded mask (adaptive percentile)
Step 3: Connected components (geometric operation)
Steps 4-6: Island analysis → entropy → metric

Introduction
This document specifies how the core kernel metrics of the Visual Thinking Lens Δx, r_v, ρ_r, μ, x□, θ, and dsare instantiated directly from gradient-field operations on an image.

The goal is to make the system's abstractions operational: each metric corresponds to a measurable transformation of the Sobel-derived pressure field, and each quantity is computed in a way that is contrast-invariant, model-agnostic, and reproducible. These metrics do not infer meaning; they measure the physical geometry of the gradient field, which is where spatial priors manifest most directly.

The approach assumes a simple but powerful premise: **an image's spatial priors are expressed most clearly in its gradient field.** Edges, voids, packing zones, and attractor lines reveal how visual mass organizes and how a model, or an artist, implicitly resolves balance, tension, and off-center pull.

By mapping each kernel metric to explicit gradient operations, this document provides a stable computational grounding for the VTL's structural analysis while preserving its theoretical framing around spatial priors, attractors, and perturbation behavior.

This is not a new vision algorithm. It is a *field translation layer*: a way of expressing compositional behavior as geometry, and geometry as measurable structure.

Spatial Priors as Forbidden Zones
The kernel metrics reveal a fundamental property of generative models: they occupy a surprisingly constrained region of compositional space.

Despite infinite prompt variations, outputs cluster around stable attractors:
- $\Delta x \in [-0.15, +0.15]$: weak center bias persists
- $r_v \in [0.30, 0.50]$: consistent void compression
- $\rho_r \in [0.20, 0.35]$: predictable packing density
- $\mu > 0.40$: rarely fragments completely

These "forbidden zones" are regions models avoid even under explicit compositional prompting. The framework detects these constraints by measuring where models *can't* go, not just where they *do* go.

Δx — Placement Offset
What it is: **Horizontal displacement of gradient mass from center.**
How computed from gradients:
1. Compute gradient magnitude field:
   $G = |\nabla I|$
2. Compute weighted centroid:
   $C_x = (\sum_x x \cdot G(x,y)) / (\sum_x G(x,y))$
3. Normalize against frame width:
   $\Delta x = (Cx - W/2)/(W/2)$

High Δx means **perceptual weight imbalance**, not subject location.
(Exactly why portraits show Δx skew even when centered.)

r_v — Void Ratio
What it is: **Percentage of the frame that contains low-pressure (low-gradient) regions.**
Gradient operation:
Let τ be a low-percentile of the gradient magnitude distribution for this image (e.g., 40th–60th percentile):
void $(p) = (G(p) < \tau)$
r_v = (# void pixels)/(# total pixels)
Using a percentile makes r_v contrast-invariant: "void" is defined relative to that image's own pressure field, not an absolute value.
This measures:
- breathing room
- compositional emptiness

- spatial calm
- negative space control

It is the **inverse** of aesthetic busyness.

ρ_r — Packing Density
What it is: **How tightly the gradient mass clusters.**
Gradient operation:
1. 1. Identify high-gradient pixels with an adaptive high-percentile threshold:
`τ_high = P_high(G),` e.g., 75th–85th percentile
`mask = G > τ_high`
2. Compute cluster density:
$\varrho\ r$ = (mass inside bounding hull)/(area of hull)
A tight cluster → high ρ_r
A diffuse field → low ρ_r
This is the metric that detects:

- tight portraits
- busy interiors
- compressed space
- visual crowding

μ — Cohesion
What it is: **Does the frame read as one subject mass or multiple fragments?**
Gradient operation:

1. Threshold gradient magnitude using the same adaptive high-percentile rule as ρ_r (high-gradient pixels = structural "islands").
2. Label connected components
3. Compute:
   μ = 1 − (entropy of island sizes)/(log K)

Here μ inherits the adaptive threshold: cohesion is measured over the dominant gradient islands for that image, not a fixed absolute scale.
High μ → one dominant structure
Low μ → fragmented or scattered

This is directly related to:

- object unity
- narrative focus
- center of attention
- structural intelligence

x_p — Peripheral Pull
What it is: **Net directional force pulling mass toward a boundary.**
Gradient operation:

1. Compute gradient-weighted centroid (as in Δx).
2. Compute distance to edges.
3. Compute pull vector:

$$x_p = \frac{1}{\sum G} \sum_p G(p) \cdot \vec{d}_{edge}(p)$$

This measures:

- edge tension
- where the image wants to "escape"
- the gravitational well of composition
- the attractor basin

This is where **your off-center prior detection** lives.

θ — Orientation Stability
What it is: **How aligned the dominant gradient directions are to the vertical axis.**
Gradient operation:
Use Sobel X and Sobel Y:
$\theta\ (p)$ = arctan $((dI/dy)/(dI/dx))$
Compute variance vs. 90° (vertical):
$\theta_{stability}$ = 1 − (Var(θ)/π^2)
High θ = stable verticals
Low θ = leaning / tilted space

ds — Structural Thickness
What it is: **Perceived depth, thickness, volumetric buildup.**
Gradient operation:
Skeletonization + dilation:

1. Skeleton shows thin structure.
2. Compare full gradient map to skeleton:
   $ds = \sum(G > \tau)\sum$(skeleton)

High ds means:

- thick forms
- volumetric light
- material presence
- painterly solidity

Low ds means:

- thinness
- graphic edges
- flattening

SUMMARY TABLE

Metric	Gradient Operation	What It Means
Δx	centroid of	∇I
r_v	% low-	∇I
ρ_r	density of high-	∇I
μ	entropy of gradient islands	cohesion vs fragmentation
x□	edge-oriented pull vector	off-center tension
θ	alignment of gradient directions	tilt vs stability
ds	ratio of gradient mass to its spine	thickness / volume

Δx — centroid of ∇I
r_v — % low-∇I
ρ_r — density of high-∇I

Implementation note: thresholds (τ, τ_high) are instantiated as percentiles of |∇I| per image, so all kernel metrics are adaptive to the image's own gradient distribution.

Conclusion

The kernel metrics defined here establish a consistent and interpretable link between an image's gradient field and its compositional behavior. By grounding Δx, r_v, ρ_r, μ, $x_{\square}$, θ, and ds in explicit, reproducible operations, the method avoids style dependence and remains stable across subjects, datasets, and generative models. This framework is intentionally model-agnostic.

It does not require training-set access, embedding introspection, or feature attribution.
Instead, it measures how spatial priors manifest in the observable structure of an image—its edges, voids, attractors, and mass distribution.

The resulting metrics serve three purposes:

1. **Comparative analysis**: distinguishing spatial biases across models and engines.
2. **Predictive steering**: quantifying how prompt forces move an image through Δx–r_v–ρ_r–μ–$x_{\square}$ space.
3. **Structural evaluation**: providing a field-based vocabulary for composition that is measurable and falsifiable.

Together, these components form a foundation for higher-level systems such as the Visual Thinking Lens, where metric behavior interacts with interpretive engines, failure analysis, and recursive scoring.
This document defines the underlying computational invariants.

Everything else in the system is built on this substrate.

Methods Summary
All metrics derive from the Sobel gradient field of a luminance-normalized image.
Preprocessing

- Convert RGB → grayscale (float32).
- Normalize to [0, 1].
- Compute gradients with Sobel operators:
  - `gx = ∂I/∂x`
  - `gy = ∂I/∂y`
- Gradient magnitude: `G = sqrt(gx² + gy²)`.

Adaptive Thresholding
Thresholds τ and τ_high are instantiated as percentiles of the gradient magnitude distribution:

- Low-percentile (e.g., 40–60) for voids (r_v).
- High-percentile (75–85) for structural mass (ρ_r, μ).

These ranges are conservative defaults suitable for most images. Specific applications may adjust:

- **Noisy/compressed images:** Widen void range (35-65) to avoid misclassifying noise as structure
- **High-contrast images:** Tighten mass range (80-90) to focus on dominant structures
- **Textured substrates:** Consider anisotropic diffusion preprocessing (see notebook Section 0)

Key principle: The ranges must remain consistent within any comparison set. Absolute values matter less than relative position in each image's own gradient distribution.

Metric Computation

- **Δx**: gradient-weighted centroid displacement from frame center.
- **r_v**: proportion of pixels with `G < τ`.
- **ρ_r**: density of high-gradient pixels within a local packing window.
- **μ**: cohesion via entropy of connected-component sizes (adaptive threshold).
- **$x_{\square}$**: peripheral pull vector computed from edge mass distribution.
- **θ**: orientation stability via arctan2(gy, gx).
- **ds**: structural thickness via ratio of gradient mass to its skeleton.

Stability
All operations are contrast-invariant and require no model internals.
The method is deterministic under fixed percentile choices and image resizing rules.

Mathematical Appendix: Formal Definitions of Kernel Metrics

Let

$$G = \sqrt{g_x^2 + g_y^2}$$

be a luminance-normalized image.

Let Sobel operators produce partial derivatives gx,gy, and define:

$$G = \sqrt{g_x^2 + g_y^2}$$

All threshold parameters τ_{low},, τ_{high} are instantiated as **percentiles** of the distribution of G over Ω.

1. Δx — Placement Offset

Define the gradient-weighted centroid:

$$C_G = \frac{\sum_{p \in \Omega} G(p)\, p}{\sum_{p \in \Omega} G(p)}.$$

Let C 0 = (W/2 , H/2) be the geometric center of the frame.

$$\Delta x = \|C_G - C_0\|_2.$$

This measures displacement of structural mass from the frame center.

2. r_v — Void Ratio

Let:

$$\tau_{\text{low}} = P_\alpha(G), \quad \alpha \in [40, 60]$$

Define void mask:

$$\text{void}(p) = \mathbf{1}\{G(p) < \tau_{\text{low}}\}.$$

Void ratio:

$$r_v = \frac{1}{|\Omega|} \sum_{p \in \Omega} \text{void}(p).$$

This normalizes void definition relative to the image's own gradient field.

3. ρ_r — Packing Density

Let:

$$\tau_{\text{high}} = P_\beta(G), \quad \beta \in [75, 85]$$

Define structural mask:

$$M(p) = \mathbf{1}\{G(p) > \tau_{\text{high}}\}.$$

Packing density is the normalized count of high-gradient pixels:

$$\rho_r = \frac{1}{|\Omega|} \sum_{p \in \Omega} M(p).$$

Alternative local formulation (optional):

$$\rho_r = \mathbb{E}_{p \in \Omega} \left[\frac{1}{|N(p)|} \sum_{q \in N(p)} M(q) \right]$$

where $N(p)$ is a local neighborhood.

4. μ — Cohesion

Let $\{C_k\}_{k=1}^K$ be connected components of the structural mask M.

$s_k = |C_k|$ and $S = \sum_k s_k$.

Define normalized island-area distribution:

$$p_k = \frac{s_k}{S}.$$

Entropy of island sizes:

$$H = -\sum_{k=1}^{K} p_k \log p_k.$$

Cohesion:

$$\mu = 1 - \frac{H}{\log K}.$$

Large μ → one dominant structure; small μ → fragmented mass.

5. x□ — Peripheral Pull

Let ∂Ω denote boundary pixels.
Define edge mass:

$$E_b = \sum_{p \in \partial\Omega} G(p)$$

and normalized boundary mass distribution:

$$w(p) = \frac{G(p)}{E_b}, \quad p \in \partial\Omega.$$

Peripheral pull vector:

$$x_p = \left\| \sum_{p \in \partial\Omega} w(p)\,(p - C_0) \right\|_2 .$$

This measures directional bias toward edges.

6. θ — Orientation Stability

Define the local gradient orientation:

$$\theta(p) = \mathrm{atan2}(g_y(p), g_x(p)).$$

Orientation stability is the concentration of the orientation distribution:

$$\Theta = 1 - \frac{H(\theta)}{\log B}$$

where $H(\theta)$ is histogram entropy over B bins.
High Θ → strong alignment; low Θ → dispersed or unstable orientation.

7. ds — Structural Thickness

Let

$$S_{\text{grad}} = \textstyle\sum_{p \in \Omega} G(p).$$

Let Γ be the morphological skeleton of G-above-threshold regions.
Define skeleton mass:

$$S_{\text{skel}} = \sum_{p \in \Gamma} G(p).$$

Structural thickness:

$$ds = \frac{S_{\text{grad}}}{S_{\text{skel}}}.$$

Large ds → thick structures; small ds → thin / filamentary mass.

Glossary

- **Δx — Placement Offset:** Distance between frame center and gradient-weighted centroid. Measures compositional displacement.
- **r_v — Void Ratio:** Fraction of low-gradient pixels. A measure of breathing room / empty space.
- **ρ_r — Packing Density:** Fraction of high-gradient pixels. Indicates compression or crowding of structure.

- **μ — Cohesion:** Entropy-normalized measure of whether structural mass forms one unified island or many fragments.
- **$x_{\square}$ — Peripheral Pull**
- Strength and direction of compositional bias toward the frame boundary.
- **θ — Orientation Stability:** Consistency of local gradient orientation; alignment vs rotational dispersal.
- **ds — Structural Thickness:** Ratio of total gradient mass to the mass of its skeleton; measures volumetric vs filamentary structure.

Attractor
A stable configuration in Δx–r_v–ρ_r–μ–$x_{\square}$ space where models repeatedly settle despite varied prompts or seeds.
Basin: A region around an attractor where small perturbations still return to the attractor state.
Perturbation: Any prompt or structural manipulation that pushes the image out of its current basin; measured by drift of kernel metrics.

Reviewer-Facing Justification
Why gradient fields are sufficient
The Sobel gradient field captures **first-order spatial structure**: edges, mass transitions, directional change, and void boundaries. Most compositional behavior—object placement, void shaping, cluster formation, edge tension—manifests first in these gradients. Unlike high-level embeddings or saliency maps:
- gradients are **model-agnostic**
- they require **no training-set access**
- they expose what is **observable**, not what is inferred

For a structural analysis framework, gradients are the minimal field that still expresses the priors a model is using.

Method	Requires Training	Model-Agnostic	Measures Geometry	Interpretable
Saliency Maps	Yes	No	No	No
Feature Attribution	Yes	No	Partial	No
CLIP Embeddings	Yes	No	No	No
Kernel Metrics	**No**	**Yes**	**Yes**	**Yes**

How this differs from saliency / feature maps
Saliency maps measure *importance* for a classifier or caption model. They reflect semantic contribution, not compositional structure. Gradient fields measure **geometry**, not meaning:
- where edges consolidate
- where voids open
- where tension accumulates
- how mass distributes across the frame

Compositional priors live here, not in semantic embeddings.This is why kernel metrics can fingerprint different image generators consistently, even when content varies.

Why adaptive thresholds are principled
Absolute gradient thresholds fail under changes in:
- contrast
- exposure
- tonal range
- subject scale

Using percentiles:
- normalizes across exposure/contrast regimes
- makes voids and mass definitions **image-relative**
- ensures r_v, ρ_r, and μ remain comparable across images

- avoids bias toward any specific luminance distribution

Percentile thresholds are standard in texture, segmentation, and robust-statistics pipelines for exactly this reason: they preserve invariance while retaining sensitivity to structure.

1. Reviewer FAQ
FAQ: Methodological Choices in Kernel Metric Computation

Q1. Why not use Laplacian-of-Gaussian (LoG) instead of Sobel?
Short answer: LoG detects *blobs* and high-frequency detail, not compositional mass.
Long answer: The Laplacian operator highlights zero-crossings and second-derivative curvature. While excellent for blob detection, LoG:

- amplifies noise in textured regions
- discards directional information
- is highly sensitive to scale selection
- does not preserve mass distribution patterns

The VTL kernel metrics require *first-order structure*: where mass lies, where voids open, and how gradient strength shifts across the frame. Sobel preserves this directly and interprets compositional tendencies without scale-to-noise amplification.

Q2. Why not use Canny edges? They're clean, stable, and popular.
Because Canny is designed for *binary contour detection*, not field analysis.
Canny imposes:

- hysteresis thresholding
- non-maximum suppression
- a thin, 1-pixel contour representation

This destroys the information we need:

- interior mass structure
- soft void boundaries
- gradient *magnitude* and *orientation* distributions
- attractor ridges across scales

Canny is excellent for object outlines; it is inappropriate for studying spatial priors, which emerge in the **continuous gradient field**, not in binarized boundaries.

Q3. Why not Difference-of-Gaussians (DoG), which approximates LoG and detects structure at multiple scales?
DoG is powerful, but not for compositional metrics. DoG:

- is tuned to detect *scale-space extrema*
- preferentially enhances blobs and texture
- suppresses planar or low-frequency compositional transitions
- produces scale-dependent responses (making Δx, r_v, ρ_r incomparable across images)

The kernel metrics require invariance to:

- scale
- contrast
- texture frequency

DoG complicates these invariances; Sobel maintains them.

Q4. Why gradient magnitude instead of raw luminance patterns?
Because composition is expressed in **changes**, not in absolute values.
Raw luminance does not tell you:

- where tension accumulates
- where voids form
- where mass consolidates

- where priors pull the structure

Gradients reveal structural hierarchy: slow changes → voids, rapid changes → mass boundaries, flows → directional bias. This makes gradients a natural basis for compositional analysis.

Q5. Why are thresholds percentile-based instead of absolute values?
Absolute thresholds fail under:

- exposure changes
- local contrast variation
- tonality shifts
- flat-color vs high-detail images

Percentile thresholds ensure that **voids and mass are defined by the image's own distribution**, not by arbitrary cutoffs. This:

- normalizes across images
- keeps metrics comparable
- avoids dataset or model bias
- stabilizes r_v, ρ_r, and μ across content types

This is a principled method used widely in robust vision pipelines.

Q6. Doesn't Sobel miss higher-level structure (objects, semantics)?
Correct, by design. The kernel operates **below semantic level**, focusing on the spatial priors expressed in gradient behavior. Semantic segmentation or saliency maps would reintroduce model biases; Sobel reveals the *observable compositional field* without interpreting content.

Q7. Why not use learned features or neural embeddings?
Because the goal is to measure **the priors expressed in the image output**, not the priors embedded in the model's weights. Learned features are:

- opaque
- architecture-specific
- dataset-dependent
- not comparable across models

Gradient fields are:

- interpretable
- deterministic
- architecture-agnostic
- stable across all generative systems

This aligns with the VTL mission: study spatial behavior, not hidden model representations.

Q8. Why is skeletonization used for ds (structural thickness)?
Skeletonization captures the "spine" of the structure independent of its thickness. Comparing gradient mass to skeleton mass yields a **scale-normalized measure of volumetric presence**, essential for distinguishing:

- filamentary forms
- planar forms
- volumetric clusters

Skeleton-based thickness avoids parameter tuning and preserves topology.

Appendix A: IEEE Kernel Metric Computation from Gradient Fields
A.1 Preprocessing
All computations operate on a luminance-normalized grayscale image

$I : \Omega \to [0, 1].$

Gradients are obtained using Sobel operators:

$$g_x = \frac{\partial I}{\partial x}, \qquad g_y = \frac{\partial I}{\partial y}, \quad G = \sqrt{g_x^2 + g_y^2}.$$

Percentile-based thresholds maintain invariance across exposure and contrast conditions.

A.2 Placement Offset (Δx)

The gradient-weighted centroid is:

$$C_G = \frac{\sum_p G(p)p}{\sum_p G(p)}.$$

Placement offset is the Euclidean distance from the geometric center:

$$\Delta x = \|C_G - C_0\|_2.$$

A.3 Void Ratio (r_v)

Let $\tau_{\text{low}} = P_\alpha(G)$, with $\alpha \in [40, 60]$.

Define:

$$r_v = \frac{1}{|\Omega|} \sum_p \mathbf{1}\{G(p) < \tau_{\text{low}}\}.$$

This measures low-gradient regions corresponding to perceptual voids.

A.4 Packing Density (ρ_r)

A high-percentile threshold identifies structural mass:

$$\tau_{\text{high}} = P_\beta(G) \ (\beta \in [75, 85])$$

$$\rho_r = \frac{1}{|\Omega|} \sum_p \mathbf{1}\{G(p) > \tau_{\text{high}}\}.$$

A.5 Cohesion (μ)

Connected components of the structural mask define island sizes sk=∣Ck∣s_k = |C_k|sk=∣Ck∣.

$$\{C_k\}_{k=1}^K$$

Normalized distribution:

$$p_k = \frac{s_k}{\sum_i s_i}.$$

Entropy-normalized cohesion is:

$$\mu = 1 - \frac{-\sum_k p_k \log p_k}{\log K}.$$

A.6 Peripheral Pull (x□)

Boundary pixels ∂Ω define the pull vector:

$$x_p = \left\| \sum_{p \in \partial\Omega} \frac{G(p)}{\sum_{q \in \partial\Omega} G(q)} (p - C_0) \right\|_2.$$

This quantifies directional bias toward edges.

A.7 Orientation Stability (θ)

Orientation:

$$\theta(p) = \operatorname{atan2}(g_y(p), g_x(p)).$$

Orientation stability is the complement of histogram entropy:

$$\Theta = 1 - \frac{H(\theta)}{\log B}.$$

A.8 Structural Thickness (ds)

Let Γ\GammaΓ be the morphological skeleton of high-gradient regions.
Define:

$$ds = \frac{\sum_p G(p)}{\sum_{p \in \Gamma} G(p)}$$

This ratio differentiates volumetric from filamentary structure.

A.9 Discussion
These metrics collectively describe spatial priors expressed in the gradient field without assuming semantic structure. Each metric is contrast-invariant, computationally lightweight, and applicable across generative model outputs.

A.10 Summary
The kernel-to–gradient-field mapping establishes a deterministic substrate for analyzing composition independently of semantics, model internals, or training data access. Each metric, Δx, r_v, ρ_r, μ, x□, θ, and ds, captures a distinct mode of spatial organization emerging directly from first- and second-order image structure. Together, they form a compact vocabulary for describing how images allocate mass, void, and directional pull, and how generative systems gravitate toward or away from specific regions of that space.

By grounding analysis entirely in the gradient field, this framework identifies **spatial priors** as measurable geometric tendencies—center bias, void compression, edge aversion, orientation stabilization, cohesion collapse, and other recurrent attractors—rather than as opaque features of latent spaces or datasets. It also reveals **forbidden zones**, compositional configurations that models rarely or never occupy even when prompted toward them, indicating strong inductive constraints in the underlying architecture.

These metrics do not claim to exhaust all dimensions of visual reasoning; instead, they define a stable, interpretable front-end that can be paired with higher-level semantic, perceptual, or generative analyses. As a result, the system functions both as an **instrument** (for diagnosing model behavior) and as a **comparative substrate** (for evaluating variation across checkpoints, engines, or human-made vs. machine-made imagery).

In short, this framework provides a reproducible way to see where compositions come from, where they resist change, and where models cannot yet go.

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Appendix B: Author Biography
Rus (A.rtist I.nfluencer / Visual Thinking Lens) is an artist and systems designer whose work bridges visual composition, machine intelligence, and computational perception. Research focuses on the gap between **human spatial intent** and **model-driven inductive bias**, specifically how images inherit structure from learned priors, and how those priors can be made observable, measurable, or disruptable.

Creator of the **Visual Thinking Lens (VTL)**, a multi-engine critique and diagnostic system integrating compositional metrics, perceptual load measurement, recursive collapse studies, and off-center prior analysis. Work combines formal art training, real-world design practice, and computational methods to build tools that expose structural tendencies in generative models without relying on semantic assumptions or training data access.

The **Kernel Metrics → Gradient-Field Operations** framework emerges from this broader agenda: to develop interpretable, deterministic instruments that describe *how images are constructed*, not what they depict. Rus’s current research centers on model fingerprinting, spatial-prior atlases, and compositional failure taxonomies for modern generative systems.

Appendix C: Vocabulary & Definitions

Compositional Kernel Metrics (Canonical)

- **Δx — Placement Offset:** Horizontal displacement of mass centroid relative to frame center. Measures off-center tendency.
- **r_v — Void Ratio:** Proportion of low-gradient area to total frame. Measures negative space and field openness.
- **ρ_r — Packing Density:** Proportion of high-gradient area to total frame. Measures compression, congestion, or material concentration.
- **μ — Cohesion:** Degree to which gradients form a unified cluster rather than multiple fragments. Derived from skeleton continuity and cluster compactness.
- **x□ — Peripheral Pull:** Net magnitude of gradient mass near frame boundaries. Indicates attraction toward edges or corners.
- **θ — Orientation Stability:** Dominant gradient orientation relative to vertical and horizontal axes. Measures tilt, rotation, or anti-gravity bias.
- **ds — Structural Thickness:** Perceived material thickness inferred from width and continuity of high-gradient strokes or forms.

Gradient-Field Terms

- **∇I (Gradient Field):** Vector field describing spatial change in luminance. Basis for all structural metrics.
- **Gradient Magnitude $|\nabla I|$:** Scalar strength of local contrast. Determines mass/void boundaries.
- **Gradient Orientation (φ):** Direction of spatial change. Used for θ, torque wheels, and flow mapping.
- **Second-Order Structure (Hessian):** Curvature and ridge detection. Identifies attractors, saddles, and basin boundaries.

Compositional Field Concepts

- **Attractor:** A stable conformation in Δx–r_v–ρ_r space toward which many images converge.
- **Forbidden Zone:** A region of composition space that generative models rarely produce, even under explicit prompting.
- **Basin:** Local low-energy region indicating a consistent spatial preference (e.g., center-left placement, upper-right lighting).
- **Perturbation Response:** How composition changes when the prompt, seed, or framing is varied.
- **Snap-Back Behavior:** The model's tendency to revert to a prior even after being temporarily pushed out of it.
- **Spatial Prior:** A statistical preference in how images arrange mass, void, and boundaries.

Appendix D: Reviewer Notes

Why Gradient Fields?

They encode local structure without semantic assumptions and are contrast-invariant. This enables a unified substrate across models, prompts, and image domains.

Why Not Saliency Maps or Attention Scores?

Those require model internals and reflect task-dependent behavior. Kernel metrics measure **native spatial organization**, not class-based relevance.

Why No Training Data Access?

Priors express themselves in output behavior. Measuring the geometry of generated images circumvents internal opacity and preserves cross-model comparability.

Why Adaptive Percentile Thresholds?

Percentile-based segmentation is invariant to exposure and global contrast, making metrics stable across photography, illustration, render engines, and AI artifacts.

Appendix E: Recommended Usage Patterns

How to use this instrument.

Model Fingerprinting

Compute Δx, r_v, ρ_r, μ across 100–1,000 samples. Compare distributions. Strong priors emerge as cluster shapes.

Comparative Basins

Map basins for GPT-Image vs. Midjourney vs. SDXL to show structural tendencies.

Prior Breach Testing

Intentionally push models toward forbidden zones (high void + high off-center) and record snap-back behaviors.

Human vs Machine Composition Studies

Compute metrics across curated artist sets vs. model images to detect systematic divergences.

Appendix F: Statement of Contribution

Contributions

1. **A deterministic mapping** from seven compositional kernel metrics to concrete gradient-field operations.
2. **A contrast-invariant analysis method** requiring no training data, embeddings, or model access.
3. **A unified diagnostic instrument** capable of evaluating spatial priors across models, prompts, and seeds.
4. **A compositional vocabulary** linking computational geometry to perceptual interpretation.
5. **A framework for identifying forbidden zones**—regions of composition space models cannot or do not occupy.
6. **A reproducible protocol** enabling cross-model comparison and regression testing for generative systems.

Authorship

This work is presented as a proposal rather than a completed proof. The methods and metrics are framed as operational scaffolds for evaluation, not as finalized benchmarks.

This framework was developed by Russell Parrish through iterative dialogue with GPT-5, Claude 3.7 Sonnet, and Gemini 2.0 Flash. The conceptual architecture, design decisions, and all substantive judgments are human-authored. The language models served as technical sounding boards, providing feedback on mathematical formulation, code implementation, statistical interpretation, and rhetorical clarity. Every structural choice originated from artistic practice and was refined through multi-model consultation.

This system was developed independently as a practitioner's tool. It does not build directly on institutional research or published critique systems but acknowledges adjacent dialogues in generative art, computational aesthetics, and perceptual theory.

This isn't a theory. It's already running.
If you're building generative tools, or trying to make them think better, this is your bridge.

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Russell Parrish. Radial Collapse Prior (RCP) A Visual Prior + Kernel-Mapped Failure Mode A.rtist I.nfluencer, 2025. ORCID: 0009-0008-9781-7995